

LINEAR REGRESSION

Linear Regression

Big complex word alert!

→ Purpose: predict a target (dependent) variable, based on one or more independent / explanatory variables.

Ex: house price prediction from house area, price square foot, nearness to main road & so on.

→ house price } target / outcome / dependent variable

→ house area, price sq foot, etc } independent variables

→ Finds the best straight line that predicts an outcome using input data.

→ LR works only when certain conditions or assumptions are met:

→ Linearity: Relationship b/w input & output is a straight line.

More study hours → better exam scores.

→ Independence: Data points don't influence each other.

→ Homoscedasticity (constant variance of errors)

→ Difference b/w. actual & predicted values should stay about the same everywhere.

→ Errors do not get bigger at higher study hours.

→ Normality of errors:

Prediction mistakes / residuals should follow a normal bell-curve distribution.

→ No strong multicollinearity:

When multiple predictors are used, - independent variables should not be closely related.

→ Ex: 'Study hours' & 'pages read' are almost the same independent variables / overlapping info.

→ No endogeneity:

→ No correlation b/w errors & predictors.

→ independent variables should not be correlated with the error term.

→ Ex: & Suppose you use 'study hours' to predict exam scores but do not include 'motivation' → which affects both study hours & scores, ∴ study hours get correlated with error, ∴ motivation X

→ Endogeneity causes biased estimates.

In short

"LR works best when the relationship is truly linear, errors are independent, & spread evenly, normally distributed & predictors are not redundant or correlated with errors."

General Equation for LR

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon$$

What we want to predict (exam score)

intercept

value of y when all predictors are 0.

(base line house price when size=0, bedrooms=0, ...)

coefficients / weights

tells us how much house price increases / when one predictor increases by 1

(sq foot inc by 1, price inc by \$200, keeping others fixed)

error / owner's negotiation, skill

Goal of LR

→ Find best values of $\beta_0, \beta_1, \beta_2, \dots, \beta_p$.

Ex: Predicting exam score y
 from study hours x_1
 ✓ no. of practice tests x_2

$$\text{Score} = 40 + 5(\text{hours}) + 10(\text{tests}) + e$$

"best coefficient values"
 $\Rightarrow$ minimizes $B_0 = 40 \Rightarrow$ base score if no study & no test taken.
 distance b/w

Predicted & actual values.
 B_1 = each study hour adds 5 points

B_2 = each practice test adds 10 points

e = leftover randomness like mood, sleep, etc.

Types

→ Simple LR: single independent var based prediction

→ Multiple LR: 2 or more ~~independent~~ var-based prediction.

Simple LR

✓ house price pred based on sq. footage only.

Multiple LR

✓ body weight pred based on height & age.

0% goal [formally]

Model fitting

→ find coeffs ($B_0, B_1, \dots, B_p$) that makes our regression line fit the data as well as possible.

→ make predicted values ($\hat{y}$) "close" to actual values (y).
 → how to measure closeness?

Residual errors! Residual = $y - \hat{y}$

→ minimize sum of squared errors:
 $\text{SSE} = \sum (y - \hat{y})^2$
 makes all errors true + penalizes large mistakes heavily

Model Evaluation

① Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

✓ avg squared difference b/w actual & predicted values!
 ✓ lower MSE = better model fit *

② R^2

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

R^2 Ranges from 0 to 1.

✓ how much of the variation in y is explained by the model
 ✓ $R^2 = 0 \Rightarrow$ Model explains nothing
 $R^2 = 1 \Rightarrow$ Perfect fit
 $R^2 = 0.8 \Rightarrow 80\%$ of the variability in scores is explained by study hours & practice tests.

How to find the Best Coefficients

① Iterative Method

→ Gradient Descent

random guesses for coeff

↓
 calculate error (MSE)

↓
 adjust coeffs step by step in the direction that reduces error

↓
 repeat until error is minimized.

↓
 follows update rule

$$B_j \leftarrow B_j - d \frac{\partial}{\partial B_j} \text{MSE}$$

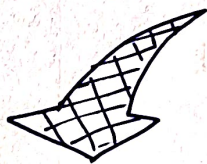
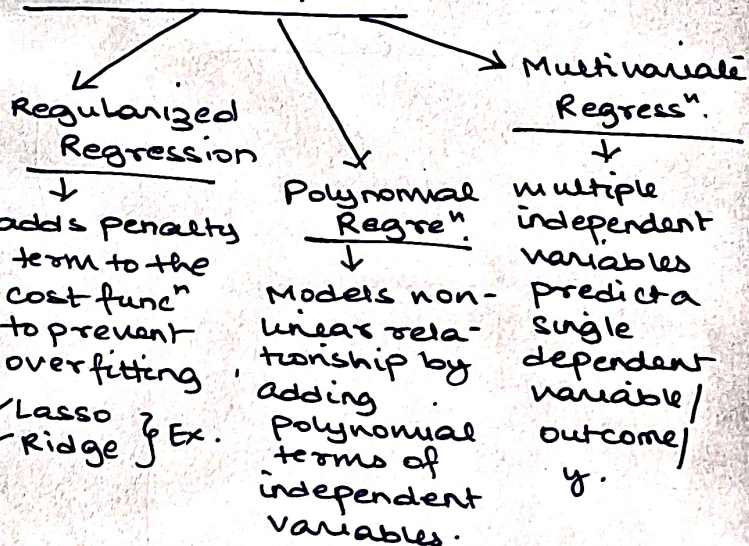
✓ for large datasets
 d = learning rate (controls step size)

② Direct Method / Non-iterative

$$\hat{B} = (X^T X)^{-1} X^T y$$

✓ for small datasets
 X = feature matrix
 y = outcome vector

Extensions of LR



Linear Regression assumes that relationship b/w features & outcome is linear, but this does not necessarily mean that the individual features themselves must always appear in the form as they are collected $\rightarrow$ without transformations.

$\hookrightarrow$ you can include non-linear transformations of the features.

Ex: $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon$
(quadratic)

or

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 x_2)$
(cross ~~features~~ features)

$\rightarrow$ Both are considered as "linear regression" as long as the parameters ($\beta_0, \beta_1, \beta_2$) are linear.

Multicollinearity - In depth

$\rightarrow$ Happens when 2 or more predictors are correlated with each other in a LR model.

✓ Overlapping, redundant info.

$\rightarrow$ why problematic?

- $\hookrightarrow$ Unstable coeff $\rightarrow$ regression cannot figure out which variable (coeff) deserves the 'credit' for y .
- $\hookrightarrow$ Interpretation difficulty.
- $\hookrightarrow$ reduced statistical power $\rightarrow$ tests become weaker $\rightarrow$ a variable that actually matters may become insignificant.

* $\rightarrow$ Inflated Variation Inflation Factor (VIF)

$\hookrightarrow$ High VIF (> 5 or 10)

signals multicollinearity

$\downarrow$
means predictor's variance is 'inflated' because of correlation with other predictors.

$\downarrow$
causes big swings in coeff values.

o.o M $\rightarrow$ limitation of LR $\rightarrow$ but not the math part! It is a data issue.

M detection $\rightarrow$ ✓ correlation matrix - close to +1 or -1.

✓ VIF $\rightarrow$ VIF > 5 or 10 .

* How to fix M?

- $\hookrightarrow$ Drop 1 correlated variable.
- $\hookrightarrow$ Combine variables.
- $\hookrightarrow$ Dimensionality reduction to new uncorrelated components.
- $\hookrightarrow$ Regularization

$\hookrightarrow$ Ridge $\rightarrow$ shrinks coeffs, reduces sensitivity to multicollinearity

$\hookrightarrow$ Lasso $\rightarrow$ shrinks coeff to 0, dropping the redundant variable.